PHYS 6031
Spring 2024
Homework 7
Due: Monday 25 March 2024 at 4:00 pm in my Clippinger mailbox

1. Consider the scalar and vector potentials $\Phi(\vec{\boldsymbol{x}}, t) = 0$ and $\vec{\boldsymbol{A}}(\vec{\boldsymbol{x}}, t) = -\frac{Ct\hat{\boldsymbol{r}}}{r^2}$, where C is a constant.

(a) Do these potentials satisfy the Lorenz condition, $\vec{\boldsymbol{\nabla}} \cdot \vec{\boldsymbol{A}} + \frac{1}{c^2}\frac{\partial \Phi}{\partial t} = 0$?
(b) Find the charge distribution.
(c) Find the current distribution.
(d) What is the gauge function $\Lambda(\vec{\boldsymbol{x}}, t)$ that will transform (i.e., via $\vec{\boldsymbol{A}}' = \vec{\boldsymbol{A}} + \vec{\boldsymbol{\nabla}}\Lambda$ and $\Phi' = \Phi - \frac{\partial \Lambda}{\partial t}$) these potentials to more conventional forms?

2. Harmonic dependence of the fields may be implmented via

$$\vec{\boldsymbol{E}}(\vec{\boldsymbol{x}}, t) = \mathrm{Re}\left[\vec{\boldsymbol{E}}(\vec{\boldsymbol{x}}) \exp(-i\omega t)\right],$$

where $\vec{\boldsymbol{E}}(\vec{\boldsymbol{x}})$ may be complex, and simmilarly for the magnetic field. Show that the time-average of the Poynting vector can be expressed as

$$\langle \vec{\boldsymbol{S}} \rangle = \frac{1}{2\mu_0} \mathrm{Re}[\vec{\boldsymbol{E}}(\vec{\boldsymbol{x}}) \times \vec{\boldsymbol{B}}(\vec{\boldsymbol{x}})^*].$$

3. A transverse plane wave is incident normally on a perfectly absorbing flat screen. Note that for a transverse electromagnetic wave the $\vec{\boldsymbol{E}}$ and $\vec{\boldsymbol{B}}$ fields are perpendicular to the direction of propagation. By assuming that linear momentum is conserved, show that the pressure (called radiation pressure) exerted on the screen is equal to the field energy per unit volume in the wave.

4. The sun has a mass of $M_\odot = 2.0 \times 10^{30}$ kg and it radiates $P = 3.8 \times 10^{26}$ W in the form of electromagnetic radiation. The gravitational constant is $G = 6.7 \times 10^{-11}$ N-m^2/kg^2.

(a) Find expressions for the gravitational and electromagnetic forces on a sphere of radius r which is located a distance R from the center of the sun , assuming that all of the electromagnetic radiation incident on the sphere is absorbed, and that the sphere is composed of material with a density of $\rho = 5$ g/cm^3.
(b) For what value of r are these forces equal? I would like for you to calculate the actual number for this part.